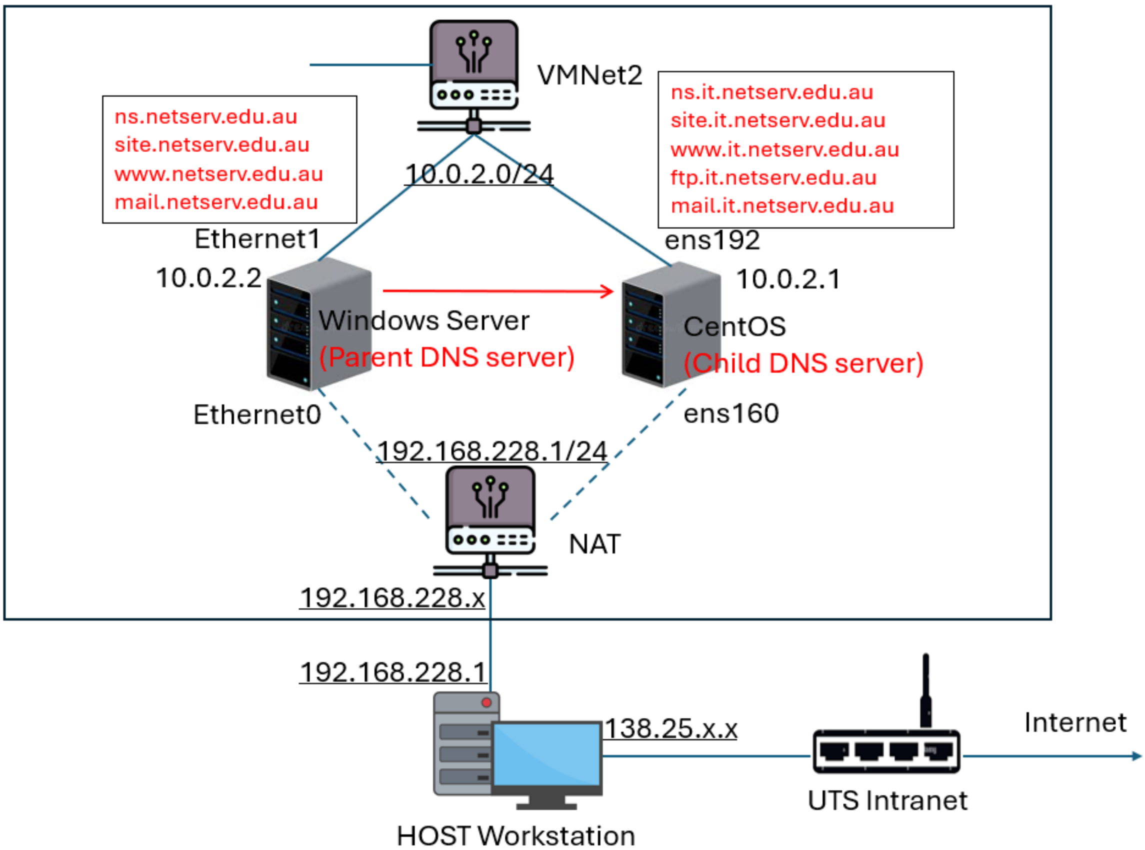
31338 Network Servers

Lab 6a: Designing and Configuring DNS Servers

Aims

1. Understand the DNS configuration
2. Configure the primary DNS service in Windows Server
3. Configure the secondary DNS service in CentOS



Task 1: Understanding the DNS Service

Before starting DNS servers, we need to understand how the two DNS services work together or independently. See the network topology as the above figure. The network settings on both systems are static. We use the second network adaptors (VMNet2) ens192 and Ethernet1 with IP addresses 10.0.2.1 and 10.0.2.2, respectively. We will set up two domain name servers. The primary DNS (Parent) will be the Windows Server, and the secondary (Child) server will be the CentOS server.

The configuration details are as follows.

DNS name	IP address	Type	Server	Notes
netserv.edu.au	10.0.2.2	NS	Windows	Primary standard zone
ns.netserv.edu.au	10.0.2.2	A	Windows	Actual Primary DNS
mail.netserv.edu.au	10.0.2.20	A	Windows	Actual Mail server
site.netserv.edu.au	10.0.2.20	A	Windows	Actual Web server
www.netserv.edu.au	10.0.2.20	CNAME	Windows	Alias to site.netserv.edu.au
it.netserv.edu.au	10.0.2.1	NS	CentOS	Secondary standard zone
ns.it.netserv.edu.au	10.0.2.1	A	CentOS	Actual Primary DNS
site.it.netserv.edu.au	10.0.2.10	A	CentOS	Actual ftp and www server
www.it.netserv.edu.au	10.0.2.10	CNAME	CentOS	Alias to site.it.netserv.edu.au
ftp.it.netserv.edu.au	10.0.2.10	CNAME	CentOS	Alias to site.it.netserv.edu.au
mail.it.netserv.edu.au	10.0.2.10	A	CentOS	Actual Mail server

We will setup both forward lookup zones and reverse lookup zones on the two servers.

Task 2: Configuring DNS Server on Windows Server

First, install the DNS role from the *Server Manager*. Once the installation is complete, in the *Tools* menu, select *DNS*. In DNS Manager, click on the server name in the left pane, and choose *Create a forward lookup zone*.

This lab will create a primary zone netserv.edu.au. Enter the zone name as netserv.edu.au. Do not allow dynamic updates in the wizard. When the forward lookup zone is set, follow the table above to set up the hosts.

Next, create the Reverse Lookup zone. Find the Reverse Lookup Zone in the left pane, and right-click it to create one. Note that the subnet is 10.0.2.0, so you only need to set the network ID 10.0.2. When the zone is created, right-click the zone file name and create a new pointer.

When the zones are set-up, restart the DNS Service.

The default name server in Windows Server is using the first network adaptor (Ethernet0, the public network). To test the localhost 10.0.2.2, you need to switch it. Open a terminal of PowerShell, type *nslookup*, and use the following commands.

```
server 10.0.2.2     (Switch to localhost)
ns.netserv.edu.au   (Test this domain name)
site.netserv.edu.au (Test this domain name)
```

The first time of lookup may be a little longer, which leads to the message `DNS request timed out`. In the answering, can the two domain names map to the IP addresses you have configured?

Task 3: Configuring BIND on CentOS 10

Linux systems implement the DNS server through the package bind. Using

```
dnf install bind
```

or

```
yum install bind
```

in the terminal to install it first.

We will use it.netserv.edu.au as the secondary zone. The name server configuration file is located at /etc/named.conf. Edit this file as follows.

1. In the "options" paragraph, find "listen-on port 53 {127.0.0.1; }; Replace 127.0.0.1 to ”any”.
2. In the same options paragraph, find "allow-query { localhost; };". Replace "localhost" to “any”.
3. Add a forward lookup zone file and a reverse lookup zone file as

```
zone "it.netserv.edu.au" IN {
       type master;
       file "it.netserv.edu.au.zone";
};
zone "2.0.10.in-addr.arpa " IN {
       type master;
       file "2.0.10.in-addr.arpa.zone";
};
```

   Note that in the above configuration, we have specified the filename it.netserv.edu.au.zone and 2.0.10.in-addr.arpa.zone in the example above for easy understanding.

The zone files are located in the /var/named/. You can now create two new zone files here by copying *named.empty* as a template. Open the zone file created in a text editor.

Hint: The SOA record entry called "rname.invalid." is an email address in a weird format. This translates to "rname@invalid", so you can change this to something like "root.netserv.edu.au.". Also note the trailing dot - it is important. The serial is currently 0, which should be incremented each time you change the DNS records.

Change the `@ IN SOA` row with the names “ns.it.netserv.edu.au. root.it.netserv.edu.au”. Then add the entries. Here is a forward zone file sample:

```
       IN     NS     ns.it.netserv.edu.au.
       IN     MX     10     mail
       IN     A      10.0.2.1
ns     IN     A      10.0.2.1
site    IN    A      10.0.2.10
www    IN     CNAME  site
```

```
ftp    IN    CNAME  site
mail   IN    A      10.0.2.10
```

The reverse zone lookup file is illustrated as follows:

```
       IN    NS     ns.it.netserv.edu.au.
10     IN    PTR    site.it.netserv.edu.au.
```

Remind the previous lab about the user account and group. The two zone files above are created by the root user, so users from other groups cannot read it. You need to use *chgrp* command to change them to *named*. After that, use *ls -l* to check that the ownership and file permissions.

The two zone files and named file are grammatically strict. Use the following commands to validate these files.

```
named-checkconf /etc/named.conf
named-checkzone it.netserv.edu.au /var/named/it.netserv.edu.au.zone
named-checkzone 2.0.10.in-addr.arpa /var/named/2.0.10.in-addr.arpa.zone
```

If you see any error messages, correct them.

The named service is blocked by the firewall by default. Use the following commands to open it.

```
firewall-cmd --add-service=dns --permanent
firewall-cmd --reload
```

Then start the service using

```
systemctl start named
```

Similar to the local test in Windows Server, you can test queries using the dig command, for example,

```
dig @localhost site.it.netserv.edu.au
dig @localhost ns.it.netserv.edu.au
dig @localhost www.it.netserv.edu.au
dig @localhost -x 10.0.2.10
```

Check the "ANSWER SECTION" of the dig output.

Also, can you try to execute the lookups using Windows Server?

Task 4: Sub-Domain Delegation

We have set up Windows Server as the parent DNS server, and CentOS as the child DNS server, respectively. How can they work together? Consider the case that when you query \`ns.it.netserv.edu.au\` from Windows server when setting up 10.0.2.2 as localhost. The name \`ns.it.netserv.edu.au\` cannot be resolved because it is configured in CentOS. This requires the delegation setting on the parent DNS server.

In the *DNS Manager*, right-click the forward lookup zone file, then choose *New Delegation*. In the wizard, input \`*it*\` as the delegated domain, then specify \`it.netserv.edu.au as\` FQDN and add the IP address 10.0.2.1. This enables the delegation.

Can you test the queries such as \`www.it.netserv.edu.au\` using 10.0.2.2 as localhost from Windows?